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From dog bite to dialysis: complement-mediated haemolytic uraemic syndrome

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SUMMARY

A previously healthy woman in her late 40s presented with rapidly progressive sepsis 4 days after a dog bite. She developed disseminated intravascular coagulation, pancytopenia and acute kidney injury, ultimately progressing to complete anuria and dialysis dependence. Blood cultures confirmed *Capnocytophaga canimorsus* bacteraemia. Despite an initial clinical improvement with antibiotics and supportive care, she developed sudden isolated haemolysis and persistent thrombocytopenia. Laboratory and renal biopsy findings confirmed the presence of a thrombotic microangiopathy (TMA), and complement-mediated haemolytic uraemic syndrome (CM-HUS) was diagnosed. Following plasma exchange and initiation of eculizumab therapy, haematological remission was achieved, and her kidney function partially recovered.

This case highlights *C. canimorsus* infection as a rare trigger of CM-HUS in immunocompetent adults and outlines the patient's diagnostic workup, the challenges of differential diagnosis and the complexity of both the clinical course and therapeutic approach in case of TMA following sepsis.

BACKGROUND

Capnocytophaga canimorsus is a slow-growing, encapsulated, gram-negative rod found in the normal oropharyngeal flora in dogs and cats. Human infections are rare, and transmission can occur via bites, scratches or even close contact. Clinical presentations are highly variable, ranging from isolated meningitis to fulminant sepsis, disseminated intravascular coagulation (DIC), multiorgan failure and thrombotic microangiopathy (TMA). Despite broad antibiotic susceptibility, mortality can approach 30%.^{1,2}

C. canimorsus is the *Capnocytophaga* species most frequently linked in invasive zoonotic disease in humans.³ Diagnosis can be delayed due to its fastidious nature and slow growth in cultures, prompting increasing use of molecular methods, such as real-time nanopore sequencing.^{4,5}

In our case, *C. canimorsus* infection presented with septic shock, severe leukopenia and coagulopathy consistent with DIC. Although the initial clinical course was dominated by sepsis-related multiorgan involvement, the later development of microangiopathic haemolytic anaemia (MAHA), thrombocytopenia and worsening acute kidney injury (AKI) suggested a superimposed TMA, highlighting how complement-mediated haemolytic uraemic syndrome (CM-HUS) may be obscured during the early sepsis and DIC phase.

CASE PRESENTATION

A previously healthy woman in her late 40s, with two prior uncomplicated pregnancies, presented to the emergency department with a rapidly worsening clinical condition. Four days prior to admission, she had been bitten by her dog on the upper lip and received tetanus vaccination as prophylaxis. She was prescribed phenoxymethylpenicillin (Apocillin) on the day of the incident; however, antibiotic treatment was not initiated, as she was unable to collect her prescription. She developed a high fever ($>40^{\circ}\text{C}$), malaise, reduced oral intake and later sought medical attention when numbness and discoloration of her feet appeared. At primary-care emergency, she was found to have hypoglycaemia, refractory to oral glucose and glucagon, with a nadir of 0.6 mmol/L, prompting urgent transfer to our emergency department.

On admission, she was alert and responsive but appeared septic with mottled and cool lower extremities. She had not used any herbal remedies or other substances, and no relatives had been ill or presented with similar symptoms. The initial laboratory workup revealed marked systemic derangement with severe inflammation, pancytopenia, AKI, liver enzyme elevation and coagulopathy. Key findings are summarised in table 1.

Blood gas analysis revealed metabolic acidosis with a pH of 7.27 (7.35–7.43) and an elevated lactate level of 5.7 (<2.5 mmol/L). She remained hypoglycaemic despite high doses of intravenous glucose, thought to be sepsis related. On arrival, the patient maintained adequate diuresis; urinalysis showed protein (2+) and blood (3+).

Given her constellation of symptoms—including fever, pancytopenia, coagulopathy with AKI and lactic acidosis—her condition was interpreted as DIC secondary to bacteraemia and early sepsis (quick Sequential Organ Failure Assessment (qSOFA) score 1/3). A surgical cause of her mottled extremities—including necrotising fasciitis—was ruled out after clinical and radiological evaluation. Due to the severity of her presentation, broad-spectrum intravenous piperacillin/tazobactam (Tazocin) was started to cover dog-associated pathogens, including *Capnocytophaga* species. Notably, *C. canimorsus* was suspected from the onset and was 7 days later confirmed by both aerobic and anaerobic blood cultures. She was subsequently transferred to the intensive care unit (ICU) for closer monitoring.

Given the pancytopenia and coagulopathy, haematology was consulted. The differential diagnoses included acute leukaemia and haemophagocytic lymphohistiocytosis. Haemolytic uraemic syndrome (HUS) was also considered but seemed



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Table 1 Laboratory profile on hospital admission

Parameter (SI units)	Patient values	Reference range
CRP (mg/L)	306	<6
Leukocytes ($\times 10^9/L$)	0.5	3.7–9.8
Thrombocytes ($\times 10^9/L$)	7	165–387
Haemoglobin (g/L)	107	117–153
Reticulocytes ($\times 10^9/L$)	27	20–100
Sodium (mmol/L)	129	137–145
Potassium (mmol/L)	3.2	3.6–4.6
eGFR (mL/min/1.73 m ²)	19	77–128
Creatine ($\mu\text{mol/L}$)	252	45–90
ALAT (U/L)	228	10–45
γ -GT (U/L)	141	10–75
Total bilirubin ($\mu\text{mol/L}$)	73	5–25
LDH (U/L)	2353	<205
Haptoglobin (g/L)	1.1	0.4–1.9
INR	2.0	0.8–1.2
APTT (s)	83	30–42
D-Dimer (mg/L FEU)	>20.0	<0.5
Fibrinogen (g/L)	1.5	2.0–4.0
Ferritin ($\mu\text{g/L}$)	39605	10–200

ALAT, *Alanine aminotransferase*; APTT, *Activated partial thromboplastin time*; CRP, *C-reactive protein*; eGFR, *Estimated glomerular filtration rate*; FEU, *Fibrinogen equivalent units*; INR, *International normalised ratio*; LDH, *Lactate dehydrogenase*; γ -GT, *Gamma-glutamyl transferase*.

unlikely due to leukopaenia, inflammatory markers, normal haptoglobin and non-compensatory reticulocytes. Flow cytometry was performed, and bone marrow biopsy planned; however, normal results—with no increase in CD34+ blasts or abnormal phenotypes—led to cancellation of the biopsy. Clinically, the presentation was consistent with sepsis. While awaiting results, supportive thrombocyte and fibrinogen transfusions were administered.

After transfer to the ICU, she became completely anuric with rapidly worsening renal function. Aggressive fluid resuscitation and concurrent administration of loop diuretics were attempted without success. She subsequently developed septic cardiomyopathy with fluid overload, requiring the initiation of continuous renal replacement therapy. Autoimmune tests (antinuclear antibodies (ANA) and antineutrophil cytoplasmic antibodies (ANCA)) were negative.

On day 3 in the ICU, she developed diffuse skin bleeding with ecchymoses and distal cyanotic and ischaemic changes affecting several toes and fingers, though pulses and sensation were still preserved. These findings were consistent with microthrombotic phenomena in the context of DIC. Despite multiorgan involvement, she never required vasopressor support beyond the stress-dose corticosteroids administered during the initial septic phase. She also received albumin infusions alongside other blood products.

After 8 days of hospitalisation, she was transferred to the nephrology ward for ongoing dialysis-dependent AKI and intravenous antibiotic therapy. On transfer, the patient was clinically stable but anuric, with generalised oedema and multiple haematomas, yet alert and oriented. Antibiotics were de-escalated to intravenous ampicillin and later to oral Apocillin based on blood culture results.

On the nephrology ward, labs showed improving inflammatory profile, reactive leukocytosis, recovering platelets and stable haemoglobin. Kidney function tests showed persistent

Table 2 Laboratory profile at admission to the nephrology ward

Parameter (SI units)	Patient values	Reference range
CRP (mg/L)	98	<6
Leukocytes ($\times 10^9/L$)	26.3	3.7–9.8
Thrombocytes ($\times 10^9/L$)	39	165–387
Haemoglobin (g/L)	111	117–153
Sodium (mmol/L)	136	137–145
Potassium (mmol/L)	3.7	3.6–4.6
eGFR (mL/min/1.73 m ²)	<15	77–128
Creatine ($\mu\text{mol/L}$)	374	45–90
Procalcitonin ($\mu\text{g/L}$)	4.07	<0.10
ALAT (U/L)	115	10–45
γ -GT (U/L)	155	10–75
Total bilirubin ($\mu\text{mol/L}$)	22	5–25
LDH (U/L)	1217	<205
INR	1.2	0.8–1.2
APTT (s)	28	30–42
Fibrinogen (g/L)	4.4	2.0–4.0
Ferritin ($\mu\text{g/L}$)	605	10–200

ALAT, *Alanine aminotransferase*; APTT, *Activated partial thromboplastin time*; CRP, *C-reactive protein*; eGFR, *Estimated glomerular filtration rate*; INR, *International normalised ratio*; LDH, *Lactate dehydrogenase*; γ -GT, *Gamma-glutamyl transferase*.

azotaemia, while coagulation parameters normalised, indicating resolving coagulopathy. Table 2 summarises her laboratory results at that stage.

After initial improvement, the clinical picture changed when she developed a sudden isolated drop in haemoglobin (111 $\rightarrow$ 68 g/L over 6 days) without evidence of bleeding or haemodynamic instability. Platelets remained persistently low ($30\text{--}40 \times 10^9/L$), haptoglobin became undetectable and LDH levels rose, indicating ongoing haemolysis (figure 1).

A direct antiglobulin test (DAT) (Coombs test) was positive for IgG, raising suspicion of autoimmune haemolytic anaemia. High-dose prednisolone was started alongside routine prophylaxis, and several blood transfusions were given over 2 days. When the haemolytic process failed to respond to corticosteroid therapy and platelets remained low ($41\text{--}46 \times 10^9/L$), further evaluation was warranted. A peripheral blood smear revealed 15%–20% schistocytes, consistent with MAHA suggestive of TMA (figure 2). The initial DAT was likely a false positive from penicillin or prior transfusions; repeat testing was negative.

Complement studies and ADAMTS13 activity were sent for analysis. Locally available standard complement tests (C3/C4) were within normal limits, and functional assays of the classical, alternative and lectin pathways were likewise normal. ADAMTS13 activity was reported shortly after, at 43.5%, effectively ruling out thrombotic thrombocytopenic purpura (TTP), as TTP typically presents with activity below 10%. While awaiting these results, plasma exchange (PEX) was initiated as bridging therapy prior to the first dose of eculizumab.

Once the thrombocyte count had increased to $77 \times 10^9/L$, a kidney biopsy was performed showing severe acute tubular necrosis, ischaemic glomeruli with fibrin thrombi and scattered crescent formation, consistent with TMA.

The diagnostic findings in this case were best demonstrated by the light microscopy features shown in figure 3. Immunofluorescence and electron microscopy are not shown, as immunofluorescence revealed no immune complex deposition and electron microscopy demonstrated no evidence of primary glomerulopathy.

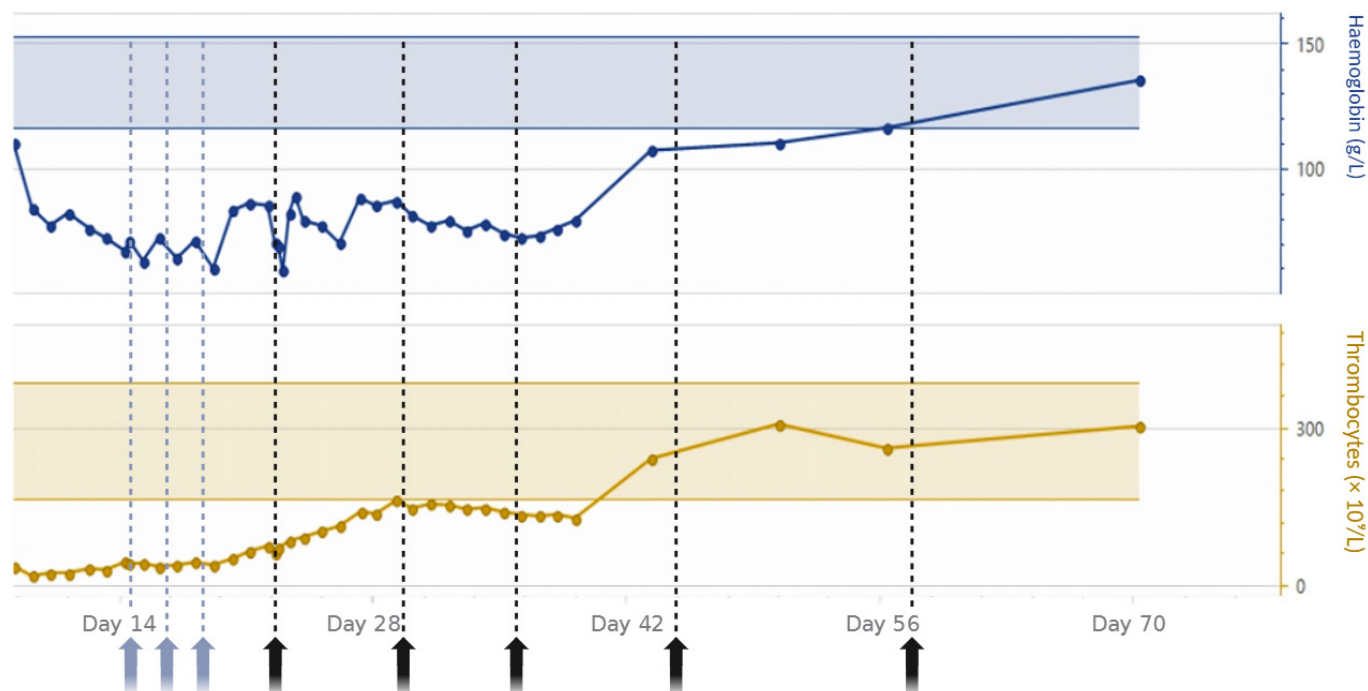


Figure 1 Evolution of haemoglobin and thrombocyte counts following onset of isolated haemolysis. Figure annotated using BioRender.com; blue arrows indicate the PEX sessions and black arrows indicate the eculizumab administration. PEX, plasma exchange.

Shiga toxin-producing *E. coli*-associated HUS (STEC-HUS) was ruled out on the basis of a negative stool PCR analysis.

Following the second and third PEX sessions, the patient's spontaneous diuresis returned. Following the first doses of meningococcal vaccines—Menveo (serogroups A, C, Y and W) and Bexsero (serogroup B)—eculizumab induction was initiated (900 mg intravenous per week for 4 weeks). The second doses of the vaccines were planned 8 weeks later, on an outpatient basis. Prophylactic Apocillin was given, as the vaccines had not been administered 2 weeks before starting complement blockade.

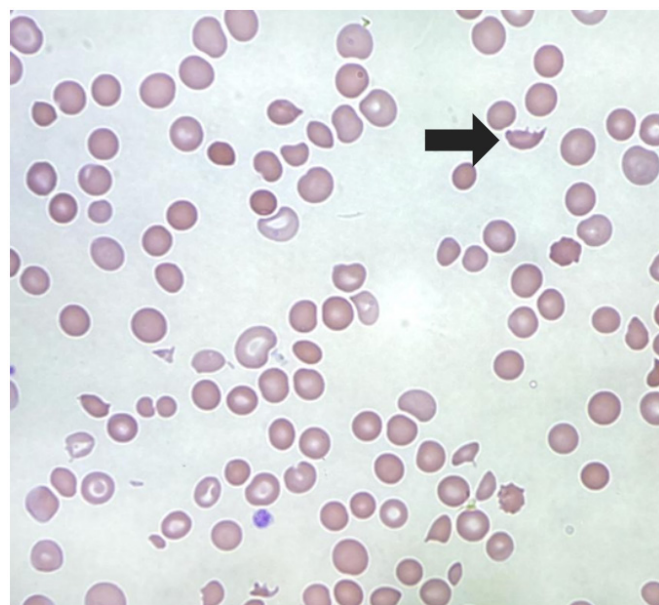


Figure 2 Peripheral blood smear showing severe schistocytosis, with a schistocyte (helmet cell) indicated by the black arrow. Occasional polychromasia and poikilocytosis are also present.

Extended complement testing to rule out hereditary complement-mediated TMA was sent to Skåne Laboratory (Lund University, Sweden). Owing to logistics and patient

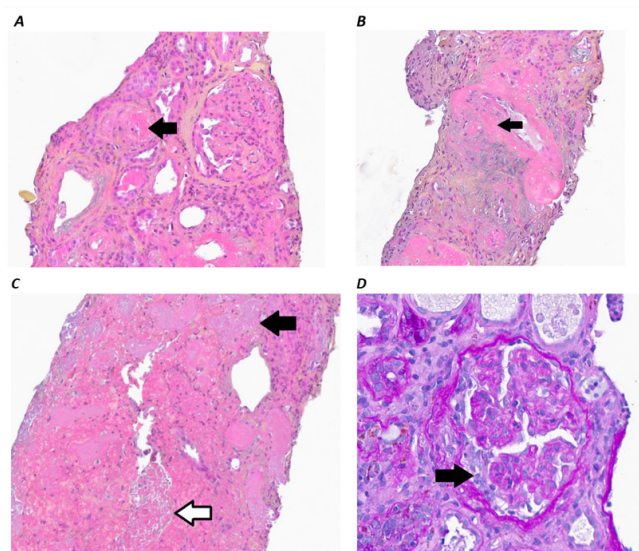


Figure 3 Kidney histopathology demonstrating TMA. (A) Arteriole with intraluminal thrombus (black arrow), consistent with acute TMA. HES stain, 20× HPF. (B) Small artery showing fibrinoid necrosis of the vessel wall (black arrow), reflecting severe endothelial injury. HES stain, 20× HPF. (C) Renal parenchyma with tubular and glomerular necrosis, and fresh interstitial haemorrhage (black arrow: necrotic tubular profile and white arrow: necrotic, blood-filled glomerulus). HES stain, 10× HPF. (D) Ischaemic glomerulus with (pseudo)cellular crescent formation (black arrow), compatible with downstream glomerular injury secondary to vascular occlusion. Periodic acid–Schiff stain, 40× HPF. HES, H&E–safranin; HPF, high-power field; TMA, thrombotic microangiopathy.

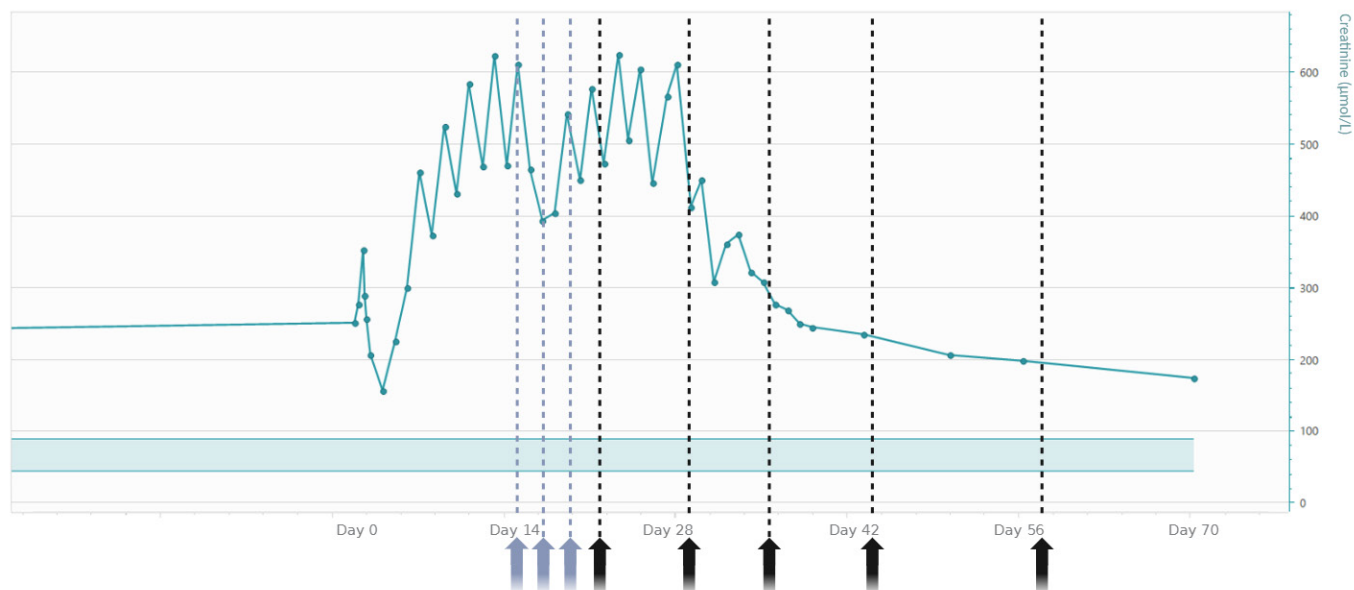


Figure 4 Serum creatine levels during hospitalisation and follow-up. Figure annotated using BioRender.com; blue arrows indicate the PEX sessions and black arrows indicate the eculizumab administration. PEX, plasma exchange.

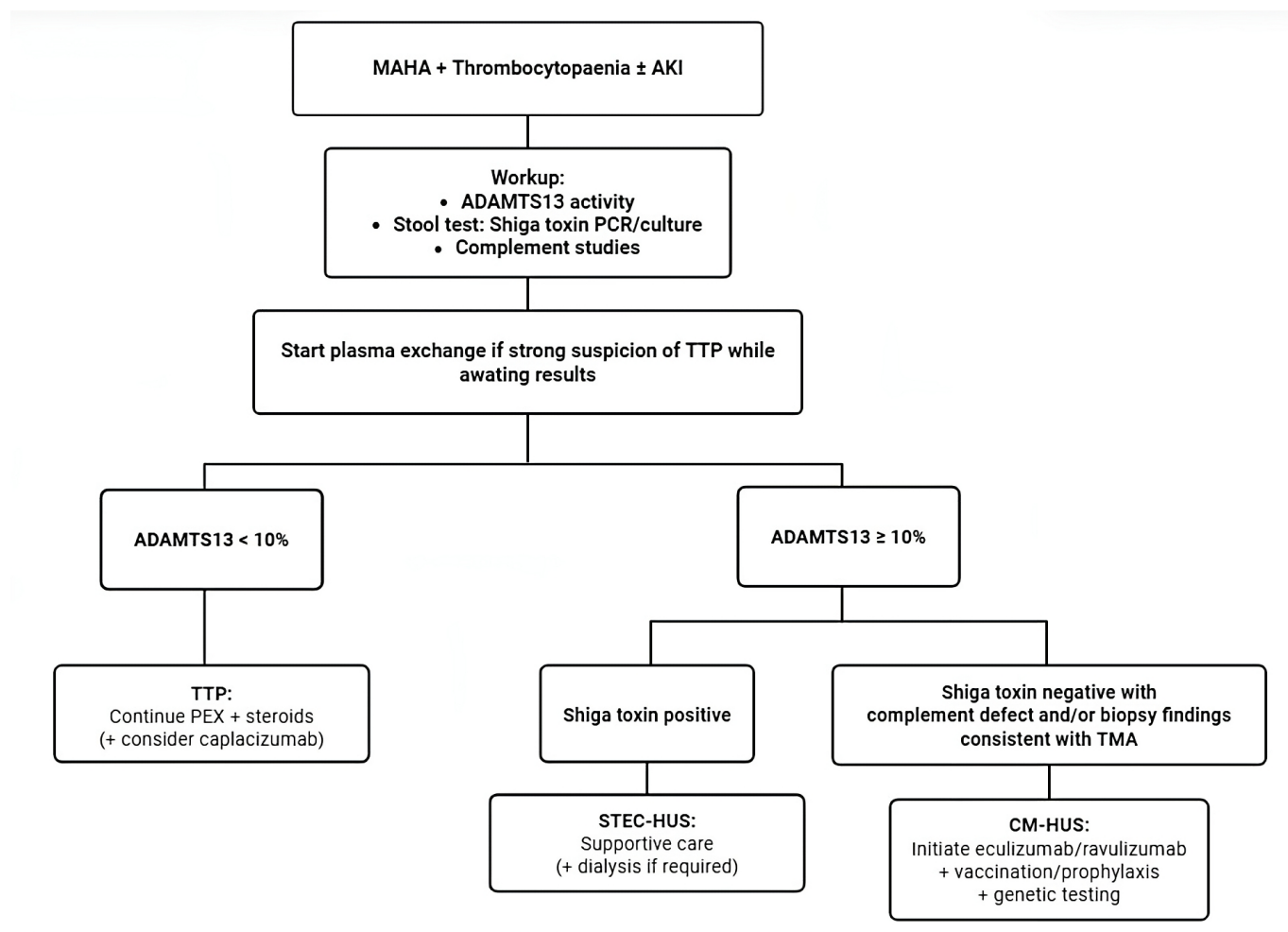


Figure 5 Proposed diagnostic algorithm for TMA. ^{13 33–35} Figure created with BioRender.com. AKI, acute kidney injury; CM-HUS, complement-mediated haemolytic uraemic syndrome; MAHA, microangiopathic haemolytic anaemia; PEX, plasma exchange; STEC-HUS, Shiga toxin-producing *E. coli*-associated HUS; TMA, thrombotic microangiopathy; TTP, thrombotic thrombocytopenic purpura.

safety considerations, three sessions of PEX with Octaplas were performed and one induction dose of eculizumab was given before sampling. Functional complement testing included alternative pathway activity (quantitative AH50), which was markedly reduced (6%), while complement factors H, I and B were normal, and antifactor H antibodies were negative.

OUTCOME AND FOLLOW-UP

Following the complex medical management, the patient is in good general condition with adequate urine output. Kidney function remains reduced (serum creatine 175 µmol/L), and she will be followed in the nephrology outpatient clinic (figure 4). She continues maintenance eculizumab (1200 mg intravenous every 2 weeks), for 4-months postinduction. Extended complement genetic testing did not identify any pathogenic variants; the results were communicated verbally by the reference laboratory, with the formal written report pending at the time of submission. Based on this information, complement blockade is being continued due to the severity of her disease, with planned reassessment for discontinuation if renal function stabilises. Identification of pathogenic complement gene abnormalities would favour longer term therapy with close monitoring for relapse.^{6 7} However, no such variants were identified in this patient.

DISCUSSION

Differential diagnosis of TMA

TMA encompasses several overlapping clinical entities.

The initial priority in adults presenting with MAHA, thrombocytopenia and AKI is to exclude thrombotic TTP, a condition defined by severe ADAMTS13 deficiency (<10%). Clinical scores like PLASMIC can help triage while awaiting the results, but diagnostic confirmation is laboratory based. Our patient's PLASMIC score was 4/7 (low pretest probability of TTP), and ADAMTS13 activity was 43.5%, making TTP unlikely and guiding further diagnostic workup.^{8–11}

STEC-HUS is another important consideration. The patient tested negative for Shiga toxin by PCR. Thus, the absence of prodromal enteric illness and negative microbiological testing argues against STEC-HUS.¹²

Ultimately, the constellation of persistent haemolysis, schistocytosis, dialysis-dependent AKI and kidney biopsy findings consistent with TMA supported a diagnosis of CM-HUS (figure 5). The diagnosis was further strengthened by the exclusion of STEC by PCR and her rapid clinical improvement after initiating eculizumab therapy. Her insignificant medical history weakens the hypothesis of a pre-existing complement defect requiring a 'second hit' and supports infection-triggered CM-HUS.

CM-HUS is a clinicopathological diagnosis of exclusion and complement levels (C3/C4) may be normal during the disease course.^{6 7} Importantly, normal complement levels (C3/C4) do not exclude CM-HUS, as levels may remain within reference ranges during the disease course. Notably, haemolysis appeared only after sepsis began improving, suggesting infection-triggered CM-HUS rather than pre-existing disease.

In our patient, basic complement tests (C3/C4) and ADAMTS13 activity were obtained immediately before the first PEX session. The extended complement testing was performed 8 days after the final PEX session and 4 days after the first dose of eculizumab. The observed reduction in the alternative pathway is most consistent with treatment effects rather than intrinsic complement depletion or activation.

Pathophysiology of CM-HUS: complement dysregulation and infectious triggers

CM-HUS (historically termed 'atypical HUS') arises from dysregulated activation of the alternative complement pathway, often in genetically predisposed individuals. Between 50% and 70% of patients harbour genetic variants or autoantibodies affecting complement regulatory proteins, such as factors H, I and B or membrane cofactor protein (CD46).^{13–15} However, disease penetrance is incomplete, and clinical manifestation typically requires an external trigger, most commonly infections, pregnancy, autoimmune disease or certain medications.^{13–19}

Although rarely reported, *C. canimorsus* can activate complement pathways, particularly in susceptible hosts. Notably, *C. canimorsus* displays variable serum susceptibility, with many isolates being complement sensitive while some capsule-expressing strains show partial resistance, which means that complement activity is an important component of host defence.^{20 21} This creates a clinical paradox: while complement activation clears the pathogen, it may also trigger uncontrolled endothelial damage and TMA in predisposed individuals.

This case adds to a small but growing body of literature linking *C. canimorsus* to complement-mediated TMAs.^{22–25} A particularly illustrative report by Björkto *et al*²² described a patient with *C. canimorsus*-induced CM-HUS harbouring a complement factor B variant, successfully treated with eculizumab. In contrast to the previously reported case, no pathogenic or likely pathogenic complement gene variants were identified in our patient, supporting infection-triggered CM-HUS without a demonstrable inherited complement predisposition.

Therapeutic strategy and clinical dilemma

Management of CM-HUS in the context of active infection presents unique challenges. Eculizumab, a monoclonal antibody targeting terminal complement component 5, is the standard of care for CM-HUS and markedly improves haematological and renal outcomes, particularly when initiated early. Delayed initiation is associated with incomplete renal recovery and consequently higher risk of progression to end-stage kidney disease.^{26–28}

In contrast, PEX is lifesaving in TTP, but provides only temporary benefit in CM-HUS, mainly by removing circulating complement fragments, inflammatory mediators or autoantibodies, without correcting the underlying complement system dysfunction.^{29 30} In our patient, diuresis returned after the final PEX sessions but before eculizumab, reflecting only transient stabilisation. Sustained haematological and renal recovery occurred only after complement blockade.

A key therapeutic dilemma is using complement blockade during active bacteraemia, as terminal complement components are critical for defence against encapsulated bacteria—including *Neisseria* and *Capnocytophaga*—raising concerns about eculizumab-associated infectious risks.^{31 32} In this case, eculizumab was initiated after blood cultures became negative, minimising this risk of uncontrolled infection. Vaccination, antimicrobial prophylaxis and close monitoring remain essential when using complement inhibitors.

CONCLUSION AND BROADER IMPLICATIONS

Early recognition of CM-HUS and prompt initiation of complement blockade are the most important predictors of favourable outcome. Our patient achieved haematological remission and partial renal recovery, consistent with the current evidence

showing that renal function and time-to-initiation of eculizumab therapy are the main determinants of prognosis.

Although *C. canimorsus*-associated CM-HUS is extremely rare, this case highlights the importance of considering infectious triggers—including unusual ones—in patients presenting with new-onset TMA. Improved awareness and molecular diagnostics may lead to the identification of more such cases.

Learning points

- *Capnocytophaga canimorsus* infection can trigger complement-mediated haemolytic uraemic syndrome (CM-HUS) in immunocompetent adults.
- Thrombotic microangiopathy may be masked during, or emerge late after, apparent sepsis recovery.
- Early initiation of eculizumab therapy improves haematological and renal outcomes in CM-HUS.
- Complement blockade should be initiated after achieving infection control.

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Contributors TEF was a part of the clinical management, collected the clinical data, performed the literature review and drafted the manuscript. B-U-DM and SD supervised the clinical management of the patient, contributed to the interpretation of findings and revised the manuscript. All authors approved the final version of the manuscript. Guarantor: SD.

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Case reports provide a valuable learning resource for the scientific community and can indicate areas of interest for future research. They should not be used in isolation to guide treatment choices or public health policy.

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